

7.4 Factoring a Trinomial With a Leading Coefficient Greater Than 1

Lesson Introduction

 Benchmark	MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real-number system.		 Learning Targets	- I can use algebra tiles to factor a trinomial with a leading coefficient greater than 1. - I can factor a trinomial with a leading coefficient greater than 1.	
 Benchmark Coherence	Building On			Working Toward	
	MA.8.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.			MA.912.AR.1.8 Rewrite a polynomial expression as a product of polynomials over the real or complex number system.	
 Essential Question	What are the similarities and differences between the process for factoring trinomials with a leading coefficient of one and a leading coefficient greater than one?				
 Lesson Narrative	In this lesson, students continue their work with factoring but will now explore factoring trinomials where the leading coefficient is not one. Students begin by connecting this process to the algebra tiles that they used in earlier lessons before moving to more abstract representations of the process. They will get the opportunity to continue practicing this process throughout the lesson.				
 Component Summary	7.4.1 Warm-Up (~5 min.)	Growth mindset activity (<i>SE p. 375</i>)	7.4.4 Try It (10 min.)	Practice factoring trinomials using preferred method (<i>SE p. 380</i>)	
	7.4.2 Guided Instruction (15 min.)	Instruction on factoring trinomials when a is not equal to one (<i>SE p. 376</i>)	7.4.5 Your Turn! (5-10 min.)	Independent practice on factoring trinomials (<i>SE p. 381</i>)	
	7.4.3 Guided Practice (10 min.)	Structured practice on factoring trinomials (<i>SE p. 379</i>)	7.4.6 Wrap-Up (~5 min.)	Muddiest Point self-reflection activity (<i>SE p. 381</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		N/A		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	Student entry points and ways of thinking may not reflect the representation of area model of factoring as presented in this lesson. For example, allow students to create their own tables in question 6d that best reflect their own way of thinking and representing their process.	- Avoid using “tricks” during this lesson. Students should see consistency through the process of factoring trinomials. They shouldn’t see factoring trinomials when $a = 1$ and when $a \neq 1$ as two different processes. Rather, use the methods previously developed in Lessons 1-3 to extend the trinomials where $a \neq 1$. - If time is an issue, consider assigning 7.4.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 7.4.2 Guided Instruction provides students with guidance on factoring trinomials where $a \neq 1$. Students will review the methods and concepts they’ve learned in prior lessons and apply them to these trinomials. As such, students will be asked to describe what they notice and differences they see as they connect what they already know to this new learning.	MA.K12.MTR.2.1: Multiple Representations This lesson provides opportunities for students to factor trinomials in multiple ways using multiple methods and approaches, such as the area model and grouping. Guide students from concrete to pictorial to abstract representations as understanding progresses. Students will continue to build understanding through modeling and using manipulatives.
 Differentiate Instruction	Notice algebra tiles as the focus of one of the lesson’s learning targets. Algebra tiles are one strategy for accomplishing the benchmark focus of rewriting polynomials, but algebra tiles are never a requirement for students to demonstrate. Continue highlighting different strategies throughout this lesson, with algebra tiles acting as one of many approaches for making sense of polynomials and factoring. - Guided Instruction Focus: Algebra tiles - Guided Practice Focus: Area model - Try It: Student choice - Your Turn!: Area model and factor by grouping	
Lesson Notes:		

7.4.1 Warm-Up: Growth Mindset

Component Overview: Students will complete a growth mindset activity by reviewing statements from celebrities and responding to growth mindset questions.



7.4.1 Warm-Up: Growth Mindset

1. Each statement describes the celebrity indicated.

Statement	Celebrity
He was never offered a top Division 1 scholarship. Scouting Reports said his explosiveness and athleticism were below standard.	Steph Curry
His first book was rejected by 27 different publishers. They said his material was too different and would never sell.	Dr. Seuss
He was fired from a newspaper for not being creative enough. They said he lacked imagination and had no good ideas.	Walt Disney

- a. How did a growth mindset help these celebrities?
Answers will vary. They turned rejection into a drive to work even harder to be successful.
- b. What can you do to maintain a growth mindset if you get frustrated with today's (or any day's) lesson to help you keep a growth mindset?
Answers will vary. I can use the frustration as fuel to be work harder to be successful with the content.



Make this more about their thinking and their discussion to these questions rather than physically writing their responses in the book. By leveraging this activity as a discussion, you are more likely to get more detailed responses from students.

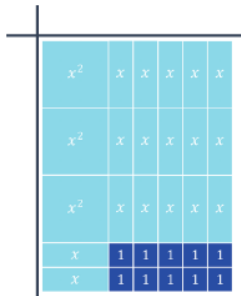
7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$

Component Overview: Students will learn how to factor trinomials where the leading coefficient is not one. They will connect their prior learning with the use of algebra tiles, the distributive property, area models, and grouping to learn how to factor this type of trinomial.



7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$

1. Use the algebra tiles to determine the two binomials that are factors of $3x^2 + 17x + 10$.



2. Discuss with your partner any differences you notice between the binomial factors in question 1 and factoring trinomials where $a = 1$. Summarize your discussion.

Answers will vary. We noticed that there was never more than one x^2 tile when $a = 1$.

3. The steps for applying the distributive property to multiply $(3x + 2)(x - 5)$ are shown.

Step	$(3x + 2)(x - 5)$	Result
Distribute $3x$ to x	$(3x + 2)(x - 5)$ 	$3x^2$
Distribute $3x$ to -5	$(3x + 2)(x - 5)$ 	$-15x$
Distribute 2 to x	$(3x + 2)(x - 5)$ 	$2x$
Distribute 2 to -5	$(3x + 2)(x - 5)$ 	-10



Guide students through this component, making explicit connections to what they have already learned when factoring trinomials where $a = 1$. Consider using Think Aloud and mark-up strategies as you model your thinking.



Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or potential misconceptions to address.



What is the relationship between the distributive property and factoring?

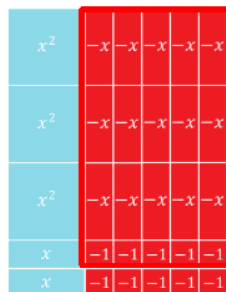
What do you notice is different about the resulting trinomial in part a from the partial products shown in the table?

7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$ (continued)

- a. Based on the results in the table, what is the product of $(3x + 2)(x - 5)$?

$$3x^2 - 13x - 10$$

- b. The algebra tiles diagram models the result of $(3x + 2)(x - 5)$.



Draw a rectangle around $-15x$ in the algebra tiles diagram for $3x^2 - 13x - 10$.

- c. Discuss with your partner how the algebra tiles diagram demonstrates the distribution of $3x$ to -5 . Summarize your discussion.

When $3x$ is distributed to -5 , the result is $-15x$. The $-15x$ tiles represent $-15x$.

4. Compare the algebra tiles diagram for $3x^2 + 17x + 10$ to the algebra tiles diagram for $3x^2 - 13x - 10$. Describe the differences.

Answers will vary. One had negative x tiles and one had positive.

Factoring trinomials of the form $ax^2 + bx + c$, where $a = 1$, resulted in the factored form $(x + g)(x + h)$, where $g \cdot h = c$ and $g + h = b$.



For question 3b, students may have a hard time connecting to the visual diagram in the book while you are providing instruction. Consider using physical tiles or a visual display of your diagram to provide a reference point while you are describing your thinking and steps.



Give students one to two minutes to discuss question 3c. Allow them the opportunity to share their thinking before moving on to question 4 in this activity. Use question 4 as an “in the moment” prompting question to elicit student responses as a whole group.



Prompt students to read and discuss this paragraph. Use mark-up strategies to highlight the connections between the form of the trinomial and its factors to lay foundations for factoring quadratic expressions in Units 8 and 9.

7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$ (continued)

5. Use the examples in the table to explore factoring trinomials of the form $ax^2 + bx + c$ where $a \neq 1$. Summarize the similarities and differences of the factored forms where $a = 1$ and where $a \neq 1$.

Trinomial	$3x^2 + 17x + 10$	$3x^2 - 13x - 10$
Factored form	$(3x + 2)(x + 5)$	$(3x + 2)(x - 5)$
Similarities	<i>Answers will vary.</i>	
Differences	<i>Answers will vary.</i>	

When factoring quadratics of the form $ax^2 + bx + c$, the coefficient b is the sum of factors of the product of a and c .

6. In $3x^2 + 17x + 10$ and $3x^2 - 13x - 10$, the products of a and c are 30 and -30 , respectively.

- a. List all the pairs of factors for both 30 and -30 .

Value	Pairs of Factors
30	1, 30; <u>2, 15</u> ; 3, 10; 5, 6
-30	$-1, 30$; $-2, 15$; $-3, 10$; $-5, 6$ 1, -30 ; <u>2, -15</u> ; 3, -10 ; 5, -6

- b. Circle the factor pair for 30 whose sum is 17.
2, 15
- c. Circle the factor pair for -30 that whose sum is -13 .
2, -15



What is the impact of having a negative value as part of this process?

Why is there double the amount of factor pairs for -30 than there is for 30? How can that play a role in the factoring process?

7.4.2 Guided Instruction: Factoring Trinomials Where $a \neq 1$ (continued)

- d. The factor pairs can be used to factor by grouping. Complete the steps for factoring by grouping for each trinomial expression.

$3x^2 + 17x + 10$	$3x^2 - 13x - 10$
$3x^2 + 2x + 15x + 10$	$3x^2 + 2x - 15x - 10$
$3x^2 + 15x + 2x + 10$	$3x^2 - 15x + 2x - 10$
$3x(x + 5) + 2(x + 5)$	$3x(x - 5) + 2(x - 5)$
$(3x + 2)(x + 5)$	$(3x + 2)(x - 5)$



Filling in the table in part d may not fit within the working schema of all students. The first line is an expansion of the expression using the factor pair. The second line is the same expansion but with the terms in a different order. This is to highlight the use of the commutative property, as students can factor using either order. In this example, the preselected work used them with the $15x$ and the $2x$. It is important for students to know that they would arrive at the same answer if they used Line 1 to factor by grouping.

7.4.3 Guided Practice: Factoring Trinomials Where $a \neq 1$

Component Overview: Students will engage in guided practice with factoring trinomials with a leading coefficient greater than one through both grouping and the use of the area model.



7.4.3 Guided Practice: Factoring Trinomials Where $a \neq 1$

- Factor $5x^2 + 6x - 8$ by grouping.
 $(5x - 4)(x + 2)$
- Consider the trinomial $3x^2 - 13x - 10$.
 - Complete the table by choosing the appropriate values.

		$3x$	☐ 1 ☒ 2 ☐ 5 ☐ 10	☐ -1 ☐ -2 ☐ -5 ☐ -10
x		$3x^2$	☐ 1x ☒ 2x ☐ 5x ☐ 10x	☐ -1x ☐ -2x ☐ -5x ☐ -10x
☐ 1 ☐ 2 ☐ 5 ☐ 10	☐ -1 ☐ -2 ☒ -5 ☐ -10	☐ 3x ☐ 6x ☐ 15x ☐ 30x	☐ -3x ☐ -6x ☒ -15x ☐ -30x	-10

- Use the values to complete the area model.

	$3x$	<u>2</u>
x	$3x^2$	<u>2x</u>
<u>-5</u>	<u>-15x</u>	-10



Allow informal observation throughout the 7.4.2 Guided Instruction to determine how “guided” this Guided Practice should be for your classroom. This could mean small-group instruction for some while others attempt independently, or it could mean teacher facilitation of practice with students actively working as a whole group. There are additional practice components to follow, so providing this additional support can solidify understanding prior to releasing students in the next component.



Students may get confused on the role of the values in the table. Consider explaining how the monomials in the middle row on the right are the product of x and the values in the column header and how the monomials in the bottom row in the middle are the product of $3x$ and the values in the row header.



Which method did you find easier to use for this trinomial? Does that preferred method match what you previously preferred to use when factoring? How can you ensure that the values maintain their correct signs?

7.4.3 Guided Practice: Factoring Trinomials Where $a \neq 1$ (continued)

3. Factor $5x^2 - 3x - 8$ using the area model.

	<u>x</u>	<u>1</u>
<u>$5x$</u>	<u>$5x^2$</u>	<u>$5x$</u>
<u>-8</u>	<u>$-8x$</u>	<u>-8</u>

$$(5x - 8)(x + 1)$$



Consider the use of a Think-Pair-Share protocol for question 3. Students should have a couple of minutes to try it on their own and then work with a partner to compare answers and discuss their processes in order to come to a consensus on the final answer.

7.4.4 Try It: Factoring Trinomials Where $a \neq 1$

Component Overview: Students will practice factoring trinomials where the leading coefficient is not one. For this activity, students can work independently or in pairs.



7.4.4 Try It: Factoring Trinomials Where $a \neq 1$

1. Factor $2z^2 - 21z - 36$.
 $(2z + 3)(z - 12)$
2. Factor $3b^2 - 23b + 14$.
 $(3b - 2)(b - 7)$



For this activity, there is no specific direction provided on the way to factor these expressions. Encourage students to use the method of their choice and find a partner who chose a different method to compare processes to make meaningful connections between representation and concept.



Consider having students factor using more than one method to add additional practice and build fluency.

7.4.5 Your Turn!

Component Overview: Students will practice factoring a quadratic expression using both the area model and by grouping.



7.4.5 Your Turn!

1. Consider the quadratic expression.

$$5x^2 + 21x + 18$$

- a. Factor using the area model.

	$5x$	6
x	$5x^2$	$6x$
3	$15x$	18

- b. Factor by grouping.

$$\begin{aligned} &5x^2 + 6x + 15x + 18 \\ &(5x^2 + 6x) + (15x + 18) \\ &x(5x + 6) + 3(5x + 6) \\ &(x + 3)(5x + 6) \end{aligned}$$



Monitor students as they work independently in this component to determine where students may need additional differentiated support.

Encourage students to use the method of their choice if they struggle with area model or factoring by grouping. If they choose a different method like algebra tiles or guess and check, pair them with a student who used area model to compare and make connections to begin building conceptual connections between methods.

7.4.6 Wrap-Up: Reflection

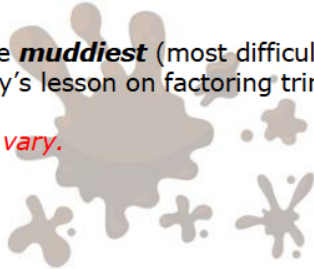
Component Overview: Students will reflect on what they found to be the most difficult or confusing point in this lesson.



7.4.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on factoring trinomials where $a \neq 1$?

Answers will vary.



*What would reflection look like for you as a teacher?
Consider your own informal data regarding students' understanding related to factoring and how to best address potential gaps in understanding throughout the remainder of the unit.*